

TRIGGER

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PROGRAM:

```
MySQL 8.0 Command Line Client

mysql> CREATE DATABASE TriggerDB;
Query OK, 1 row affected (0.04 sec)

mysql>
mysql> USE TriggerDB;
Database changed
mysql>
mysql> CREATE TABLE STUDENT (
  -> ID INT PRIMARY KEY,
  -> Name VARCHAR(100),
  -> Email VARCHAR(100)
  -> );
Query OK, 0 rows affected (0.10 sec)

mysql>
mysql> INSERT INTO STUDENT (ID, Name, Email)
  -> VALUES
  -> (1, 'Harry Potter', 'harry@gmail.com'),
  -> (2, 'Clark Kent', 'clark@gmail.com'),
  -> (3, 'Tony Stark', 'tony@gmail.com');
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> SELECT * FROM STUDENT;
+----+-----+-----+
| ID | Name   | Email      |
+----+-----+-----+
| 1  | Harry Potter | harry@gmail.com |
| 2  | Clark Kent  | clark@gmail.com |
| 3  | Tony Stark  | tony@gmail.com  |
+----+-----+-----+
3 rows in set (0.01 sec)

mysql>
mysql> CREATE TABLE STUDENT_AUDIT (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> student_id INT,
  -> lastname VARCHAR(100),
  -> email VARCHAR(100),
  -> changedat DATETIME,
  -> action VARCHAR(20)
  -> );
Query OK, 0 rows affected (0.04 sec)
```

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+----+-----+-----+
| 1  | Harry Potter | harry@gmail.com |
| 2  | Clark Kent  | clark@gmail.com |
| 3  | Tony Stark  | tony@gmail.com  |
+----+-----+-----+
3 rows in set (0.01 sec)

mysql>
mysql> CREATE TABLE STUDENT_AUDIT (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> student_id INT,
  -> lastname VARCHAR(100),
  -> email VARCHAR(100),
  -> changedat DATETIME,
  -> action VARCHAR(20)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER before_student_update
  -> BEFORE UPDATE ON STUDENT
  -> FOR EACH ROW
  -> BEGIN
  -> INSERT INTO STUDENT_AUDIT
  -> SET
  -> student_id = OLD.ID,
  -> lastname = OLD.Name,
  -> email = OLD.Email,
  -> changedat = NOW(),
  -> action = 'update';
  -> END //
Query OK, 0 rows affected (0.03 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> UPDATE STUDENT
  -> SET Name = 'Tony Stark C',
  ->     Email = 'tonystark@gmail.com'
  -> WHERE ID = 3;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```



OUTPUT:

```
MySQL 8.0 Command Line Client
mysql> DELIMITER ;
mysql>
mysql> UPDATE STUDENT
  -> SET Name = 'Tony Stark C',
  ->     Email = 'tonystark@gmail.com'
  -> WHERE ID = 3;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM STUDENT;
+----+-----+-----+
| ID | Name   | Email           |
+----+-----+-----+
| 1  | Harry Potter | harry@gmail.com |
| 2  | Clark Kent  | clark@gmail.com  |
| 3  | Tony Stark C | tonystark@gmail.com |
+----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM STUDENT_AUDIT;
+----+-----+-----+-----+-----+-----+
| id | student_id | lastname | email           | changedat | action |
+----+-----+-----+-----+-----+-----+
| 1  |          3 | Tony Stark | tony@gmail.com  | 2026-09-01 08:12:48 | update |
+----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql>
```

